AMBIENCE HEALTHCARE ENGINEERING

# Backend Transcription Take-Home Prompt

## 1. Task Overview

Build an audio transcription service that transcribes audio files of patient/provider conversations and returns transcripts. Clients submit jobs containing chunks of audio to be transcribed and stitched into a single transcript. Build this **as if you were shipping it to production at Ambience**.

## 2. System Constraints & ASR Characteristics

**ASR Chunk Limit**

Processes audio chunks ≤ 5 minutes in length. Input chunks are guaranteed to be ≤ 5 mins.

**Latency Expectation**

Each ASR call has 5–10s latency. Target full transcript return within <20s in the happy path.

**Flakiness & Failures**

Anticipate 1/20 requests failing transiently. Some audio segments will permanently fail.

**Strict Concurrency Cap**

Enforce ≤ 100 simultaneous ASR requests. Exceeding risks vendor bans or financial penalties.

## 3. Functional Requirements (API Contract)

```
POST /transcribe
  input: { audioChunkPaths: string[], userId: string }
  returns: { jobId: string }

Type TranscriptResult = {
    transcriptText: string,
    chunkStatuses: { [audioPath: string]: chunkStatus },
    jobStatus: string,
    completedTime: DateTime/ISOString
}

GET /transcript/{jobId}
  returns: TranscriptResult

GET /transcript/search
  queryParams: jobStatus?: string, userId?: string
  returns: TranscriptResult[]
```

## 4. Non-Functional Requirements

- **Fault Tolerance**: Jobs resume gracefully after server/worker restarts via persistent state.
- **Concurrency Adherence**: Strict ≤ 100 simultaneous active ASR requests across all workers.
- **Latency Target**: Target <20s happy-path latency using parallel chunk dispatch.
- **Scalability**: Architecture must scale cleanly from 1,000 to 10,000 DAU without redesign.

## 5. Scale & Workload Datapoints

- **Scale Today**: 1,000 Daily Active Users (DAU), 8-hour shifts, 30-minute visits.
- **Growth Forecast**: Expected to scale 10x to 10,000 DAU within 6 months.
- **Encounter Duration**: Usually 30 minutes, with spikes up to 120 minutes.
- **Audio & Speech Metrics**: ~140 words per minute; audio files ~1 MB per minute.

## 6. Pre-Submission Q&A Clarifications

**Deployment & Auth**

Deployment is out of scope (local execution is fine). Authentication is also out of scope.

**Search Parameters**

Both `jobStatus` and `userId` query parameters are optional. Searches span across all time.

**Client Behavior**

Assume well-behaved clients sending optimal audio chunks (no spam or malformed segment structures).

## 7. Deliverables & Time Budget

- **Time-box**: ~8 hours. AI development tools (Copilot, Cursor, Claude Code) are encouraged.
- **Submission**: Push completed code to the assigned private GitHub repository 24h prior to presentation.
- **Presentation**: 1-hour Google Meet walk-through including code demo, IDE/terminal access, and system design diagram review.

## 8. Starter Code & Reference Links

**ASR Mock Service Repository**:
https://github.com/schwers/asr-sample-service/blob/main/README.md